Cell Structure

Learning objectives

- Identify the main organelles found in eukaryotic cells and state their functions.
- Compare prokaryotic and eukaryotic cells.
- Recognise structural features unique to plant cells.
- Explain why cell structure underpins how cells generate energy.

Explanation

All living organisms are composed of cells, and their internal organisation determines what a cell can do. Cells fall into two broad categories: prokaryotic cells (such as bacteria), which lack a nucleus and instead keep their DNA within a region called the nucleoid, and eukaryotic cells (such as animal and plant cells), which contain a nucleus and various membrane-bound organelles. The nucleus is enclosed by a double membrane and holds the cell's chromosomal DNA; this DNA contains genes whose instructions influence the cell's activities. Mitochondria, also double-membraned, carry out many of the reactions involved in aerobic respiration and use energy released from glucose to produce ATP, though this is only part of the overall respiration process. Ribosomes, either free-floating or attached to the endoplasmic reticulum, assemble proteins. The cell membrane, a selectively permeable phospholipid bilayer, regulates the movement of substances into and out of the cell.

Key points

- Prokaryotic cells lack a nucleus, with DNA located in the nucleoid region; eukaryotic cells have a nucleus enclosed by a double membrane.
- Mitochondria carry out many reactions of aerobic respiration and use energy released from glucose to produce ATP, but are not the sole site of respiration.
- Ribosomes build proteins from genetic instructions.
- Many mature plant cells have a large permanent vacuole, and a cellulose cell wall; chloroplasts occur mainly in photosynthetic tissues, so not every plant cell contains them.
- The cell membrane controls what enters and leaves the cell.

Example

Cells with high energy demands, such as many muscle cells, usually contain numerous mitochondria because they require a continuous supply of ATP.

Why this matters

Knowing which structures exist, and where, explains why respiration can involve multiple cellular locations rather than a single organelle — a distinction explored in the next section.

Quick check

Name one structural difference between a prokaryotic and a eukaryotic cell, and explain why mitochondria alone don't account for all stages of respiration.

Cellular Respiration

Learning objectives

- Describe the overall purpose and general equation of aerobic respiration.
- State where glycolysis and the later stages of aerobic respiration take place.
- Compare aerobic respiration with anaerobic pathways.
- Explain how temperature affects respiration rate.

Explanation

Building on the organelles introduced in the previous section, respiration is the process by which cells break down glucose to release energy. Aerobic respiration can be summarised as glucose + oxygen → carbon dioxide + water, with energy transferred to ATP. It begins with glycolysis, which takes place in the cytoplasm and does not require oxygen. Glycolysis splits glucose into smaller molecules, yielding a small amount of energy. If oxygen is available, the products of glycolysis, including pyruvate, can enter the mitochondria, where the later stages of aerobic respiration occur, releasing far more energy overall. Aerobic respiration depends on a continuous oxygen supply to proceed, although oxygen is used directly only at a specific point within these later stages rather than throughout every reaction. When oxygen is limited or absent, cells rely instead on anaerobic pathways. In animal cells this takes the form of lactic acid fermentation; in yeast, it produces ethanol and carbon dioxide, known as ethanol fermentation. Both anaerobic routes release considerably less energy than the aerobic pathway. Because respiration is controlled by enzymes, its rate generally increases as temperature rises towards an optimum, but falls at higher temperatures when enzymes begin to lose their functional shape.

Key points

- Glycolysis occurs in the cytoplasm and does not need oxygen.

- Later aerobic stages occur mainly in the mitochondria described earlier.
- Aerobic respiration depends on oxygen overall, though not every individual stage uses it directly.
- Animal cells use lactic acid fermentation; yeast uses ethanol fermentation.
- Respiration rate generally rises with temperature up to an optimum, then may decline as enzymes lose function.
- Aerobic respiration can be summarised as: glucose + oxygen → carbon dioxide + water, with energy transferred to ATP.

Example

During intense exercise, muscle cells may temporarily switch to lactic acid fermentation when oxygen delivery cannot keep pace with demand.

Why this matters

Respiration is only useful to a cell because it produces energy in a usable form. The next section examines how that energy is actually captured and transferred to cellular processes using ATP.

Quick check

Explain why glycolysis can continue even when oxygen is unavailable, but the later stages of aerobic respiration cannot.

Energy Transfer in Cells

Learning objectives

- Describe the roles of ATP and ADP in cells.
- Explain how ATP hydrolysis releases energy for cellular use.
- Identify processes that rely on ATP as an energy transfer molecule.
- Explain why some energy is always lost as heat during these transfers.

Explanation

Cells use much of the energy released during respiration to add a phosphate group to ADP, producing ATP. ATP acts as a short-term energy carrier, transferring energy in small, manageable amounts to processes that require it. When a cell requires energy, ATP is hydrolysed back into ADP and phosphate; overall, this hydrolysis reaction releases usable energy, which the cell can then couple to other processes. It isn't simply that breaking a chemical bond releases energy on its own — the energy comes from the overall change between the reactants and products of the reaction. Because ATP is broken down and reformed continuously rather than stockpiled, it functions as an energy transfer molecule rather than a long-term energy store; longer-term energy stores include glycogen and lipids.

Key points

- ATP is formed by adding a phosphate group to ADP, using energy released during respiration.
- Hydrolysis of ATP back to ADP and phosphate releases usable energy overall.
- ATP transfers energy for immediate use rather than storing it long-term.
- Active transport, muscle contraction, and biosynthesis all rely on energy transferred via ATP.
- Energy transfers are not completely efficient, so some energy is dissipated to the surroundings as heat.

Example

During muscle contraction, ATP hydrolysis provides the energy that allows protein filaments within muscle fibres to slide past one another.

Why this matters

Because ATP links respiration to many energy-requiring activities in a cell, understanding it explains how the chemical energy released in the previous section actually gets put to use.

Quick check

Explain why ATP is described as an energy transfer molecule rather than an energy storage molecule.

DNA Structure

Learning objectives

- Describe the structure of a DNA nucleotide and the double helix.

- Explain complementary base pairing and its role in accurate replication.
- Distinguish between DNA, genes, and chromosomes.
- Recognise that not all DNA sequences function as protein-coding genes.

Explanation

DNA (deoxyribonucleic acid) is the molecule that carries genetic information in almost all living organisms. It's built from repeating units called nucleotides, each consisting of a sugar molecule, a phosphate group, and one of four nitrogenous bases: adenine, thymine, guanine, or cytosine. The sugar and phosphate groups of neighbouring nucleotides link together to form a sugar-phosphate backbone, giving each DNA strand a stable structural framework. Two such strands wind around each other to form the double helix, held together by hydrogen bonds between bases on opposite strands. These bases pair in a fixed pattern, known as complementary base pairing: adenine always pairs with thymine, and guanine always pairs with cytosine. This predictable pairing means that each existing strand can act as a template for building a new complementary strand, allowing the DNA sequence to be copied accurately when cells divide. A chromosome consists of a long DNA molecule associated with proteins; in eukaryotic cells, it becomes highly condensed during cell division, and a gene is a section of DNA containing instructions for producing a functional product, such as a protein or an RNA molecule. However, not every stretch of DNA along a chromosome functions as a gene — much of it plays other roles or has no currently identified function.

Key points

- Nucleotides consist of a sugar, a phosphate group, and a nitrogenous base.
- Complementary base pairing (A–T, G–C) allows accurate copying of genetic information.
- A chromosome consists of a long DNA molecule associated with proteins; a gene is a section of DNA containing instructions for a functional product.
- Not all DNA sequences are protein-coding genes.

Example

If one DNA strand reads A-T-G-C, the complementary strand must read T-A-C-G, allowing each strand to serve as a reliable template for copying.

Why this matters

Understanding how DNA is physically organised, and that only certain sections function as genes, sets up the next section, which explains how the information within genes is actually used by the cell.

Quick check

Explain how complementary base pairing helps ensure that DNA is copied accurately.

Gene Expression

Learning objectives

- Describe transcription and translation, and state where each occurs.
- Explain the roles of mRNA, ribosomes, codons, and amino acids.
- Recognise that genes can produce proteins or functional RNA molecules.
- Understand that different cells can activate different genes.

Explanation

The sections of DNA introduced in the previous section only have an effect on a cell once their information is put to use, through a process called gene expression. This happens in two main stages. The first, transcription, takes place in the nucleus, where the DNA sequence of a gene is used as a template to build a single strand of messenger RNA (mRNA). This mRNA then moves out of the nucleus into the cytoplasm, where the second stage, translation, occurs at ribosomes in the cytoplasm or on the rough endoplasmic reticulum. During translation, the ribosome reads the mRNA sequence in groups of three bases called codons; most codons specify an amino acid, while some act as stop signals that mark the end of translation. Transfer RNA molecules carry specific amino acids to the ribosome, where their anticodons pair with complementary codons on the mRNA. As the ribosome moves along the mRNA, amino acids are joined together in the order specified, building a protein. Many genes are expressed in this way to produce proteins, but not all of them are: some genes are transcribed into RNA molecules that carry out a function directly, without ever being translated into protein. Which genes a cell actually expresses depends on its type and role — a nerve cell and a liver cell usually contain the same set of genes but activate different combinations of them, producing very different characteristics and functions.

Key points

- Transcription (DNA to mRNA) occurs in the nucleus; translation occurs at ribosomes in the cytoplasm or on the rough endoplasmic reticulum.
- Most codons specify an amino acid, while some act as stop signals.
- Amino acids are joined in sequence during translation to form a protein.
- Some genes produce functional RNA molecules rather than proteins.

- Different cell types usually contain the same set of genes but activate different combinations of them.

Example

The insulin gene is actively expressed in pancreatic beta cells but remains switched off in most other cell types.

Why this matters

Gene expression helps explain how genetic information leads to observable characteristics, which is a useful, though not always essential, foundation for understanding the inheritance patterns covered next — many basic inheritance examples can be understood through allele combinations alone.

Quick check

Explain the difference between transcription and translation, including where each takes place.

Genetic Inheritance

Learning objectives

- Define allele, genotype, and phenotype.
- Explain dominant and recessive inheritance using a Punnett-square example.
- Distinguish diploid body cells from haploid gametes.
- Recognise that many traits involve more than simple single-gene inheritance.

Explanation

Genes, as introduced previously, can exist in alternative versions called alleles. Most human body cells are diploid, meaning they contain two sets of chromosomes, one inherited from each parent. For most autosomal genes, this gives the cell two alleles for each gene. Gametes (sperm and egg cells), by contrast, are haploid, containing only one copy of each gene, so that when they combine at fertilisation, the resulting offspring receives the usual two copies. An organism's genotype is the specific combination of alleles it carries for a gene, while its phenotype is the observable characteristic produced by the interaction between its genotype and the environment. In simple cases of complete dominance, a dominant allele affects the phenotype when one or two copies are present, while the recessive phenotype appears only when two recessive alleles are inherited. Consider a plant where the allele for purple flowers is dominant and the allele for white flowers is recessive: crossing two plants that each carry one of each allele would be expected to produce offspring in a ratio of roughly three purple-flowered to one white-flowered, though this reflects probability across many offspring rather than a guaranteed outcome for any single individual. Many characteristics don't follow this simple pattern at all — traits such as height or skin colour are often polygenic, influenced by multiple genes together with environmental factors. Mutations can arise through errors during DNA replication or through DNA damage caused by environmental factors, creating new alleles.

Key points

- Alleles are alternative versions of a gene; genotype is the allele combination, phenotype is the observable trait.
- Most body cells are diploid, containing two sets of chromosomes; for most autosomal genes, this provides two alleles; gametes are haploid (one allele copy).
- In simple cases of complete dominance, one dominant allele can affect the phenotype, while the recessive phenotype requires two recessive alleles.
- Predicted inheritance ratios describe probability across many offspring, not a certainty for any one individual.
- Many traits are polygenic or environmentally influenced, and mutations can generate new alleles.

Example

Crossing two pea plants that each carry one dominant (tall) and one recessive (short) allele would be expected to produce roughly three tall plants to every one short plant among their offspring, on average.

Why this matters

Understanding how alleles combine and separate explains observable patterns of inheritance across generations, building on the genetic groundwork laid by DNA structure and gene expression.

Quick check

Explain why an individual offspring might not match the "expected" ratio predicted by a Punnett square, even though the ratio itself is statistically accurate.

Historical Biology Experiments

Learning objectives

- Describe Robert Hooke's observations of cork and their significance.
- Explain what Mendel's pea-plant experiments revealed about inheritance.
- Describe how the Hershey–Chase experiment identified DNA as genetic material.
- Explain the contributions of Rosalind Franklin, and Watson and Crick, to determining DNA's structure.

Explanation

Much of the biology covered so far was established gradually, through the work of many researchers over several centuries, rather than through a single continuous line of discovery. In the seventeenth century, Robert Hooke examined thin slices of cork using an early compound microscope and observed small, box-like compartments, which he named "cells" — an early observation that eventually contributed to the broader development of cell theory. Two centuries later, Gregor Mendel bred pea plants and tracked how characteristics such as flower colour and seed shape were passed between generations, noticing predictable ratios that suggested inheritance followed identifiable rules, long before genes or DNA were understood. In 1952, the Hershey–Chase experiment used radioactively labelled bacteriophage viruses to demonstrate that DNA, rather than protein, enters bacterial cells during bacteriophage infection and carries the information needed to produce new viruses. Around the same period, Rosalind Franklin and her research student Raymond Gosling produced detailed X-ray diffraction images of DNA. Their evidence, including Photograph 51, revealed important features of DNA's helical structure and dimensions that proved essential to working out DNA's shape. James Watson and Francis Crick then proposed the double-helix model of DNA in 1953, drawing on Franklin's imaging data alongside chemical evidence gathered by other researchers, rather than arriving at the structure through their own experimental work alone.

Key points

- Hooke's observation of cork gave rise to the term "cell."
- Mendel's pea-plant experiments revealed predictable patterns of inheritance.
- The Hershey–Chase experiment showed that DNA, not protein, carries the genetic information passed on during bacteriophage infection.
- The X-ray diffraction work of Franklin and Gosling supplied key structural evidence for DNA.
- Watson and Crick's model relied on evidence contributed by multiple researchers.

Example

Photograph 51, produced through the X-ray diffraction work of Franklin and Gosling, provided important evidence for DNA's helical structure.

Why this matters

These experiments illustrate how the concepts explored elsewhere in this guide were established through evidence gathered by many different scientists, offering useful context rather than a required step before studying any other topic.

Quick check

Explain why it would be inaccurate to describe the discovery of DNA's structure as the work of only one or two scientists.